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VeerDrishti: A Computer Vision Framework for Real-Time Situational Awareness and Visual Threat Assessment

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Abstract

In high-stress combat environments, operators must process a lot of visual information while dealing with constant positioning, movement, poor visibility, stress and mental fatigue. Continuous manual monitoring makes it difficult to spot threats, targets and points of interest quickly. It can slow decision-making leading to operational difficulties and casualties. VeerDrishti is an AI-powered computer vision framework that aims to provide real-time situational awareness and automated visual threat assessment. The system captures live video using a camera module and processes each frame using a on-device detection model. The detection pipeline does frame pre-processing, real-time inference and post-processing before it shows the detected objects with bounding boxes, labels and confidence scores on an augmented reality interface. The framework focuses on low-latency processing so that visual detection happens directly on edge hardware without an external computing infrastructure. So far tests have shown real-time operation at 18-25 frames per second with low latency on inexpensive low-end hardware under different conditions. Further evaluation on field conditions has to be done to assess accuracy, robustness and reliability in real-world operational environments.

Keywords: *Artificial intelligence, computer vision, real-time inference, object detection, classification, low-latency, edge deployment, augmented reality*